

From single- to multi-ancestry fine-mapping at scale: a Gentropy workflow

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Abstract

Fine-mapping resolves which of the correlated variants at an associated GWAS locus drives a phenotype, assigning each variant a posterior inclusion probability and aggregating them into a credible set that captures the true signal at a defined posterior probability. The Open Targets Gentropy framework already applies eight ancestry-specific fine-mapping strategies at scale, incorporating the PICS and SuSiE-inf approaches, and continuously fine-maps GWAS across the Open Targets Platform — one study at a time, within a single ancestry. The same resource holds many GWAS for the same trait across different ancestries. Grouping GWAS Catalog studies in Open Targets release 26.06 by shared trait yielded 7,490 trait-sets containing multiple single-ancestry studies. A variant that tags the true signal through linkage disequilibrium, without driving it, shows different effects across populations because LD structure differs; a tag statistically indistinguishable from the true driver in one ancestry can be distinguishable in another. Multi-ancestry fine-mapping exploits these differences by integrating data across populations, improving resolution for causal variants shared across ancestries. Moving from single-study, single-ancestry fine-mapping to a multi-ancestry, multi-trait approach is therefore the logical next step. We are developing a scalable multi-ancestry fine-mapping workflow that integrates GWAS summary statistics with ancestry-matched linkage disequilibrium reference panels from gnomAD and Pan-UKBB. The workflow is implemented using Nextflow on top of the Open Targets Gentropy framework and is designed to run in-house at biobank scale. The pipeline resolves the Gentropy study index; for each trait-set it determines which combinations of studies are eligible for multi-ancestry fine-mapping and routes each eligible combination to one or more applicable methods. Studies assigned to an execution route are clumped, and loci with their matched LD are passed through a shared Gentropy input interface. The method itself is agnostic and selectable, spanning MultiSuSiE as proof of concept, with future work planned to integrate other methods. Method output is normalised through a shared Gentropy output interface back into the standard credible-set format, so it can be reused by other post-GWAS analysis methods. This workflow will allow Open Targets to produce more accurate credible sets at scale. Built on open-source methods and uniform access to external

LD reference panels, it will support reuse by other groups working with multi-ancestry GWAS data.